

Programming Constructs Loops For - arrays Foreach - collections While - wait for a condition do-while - runs at least once Increment/Decrement Operators a++ / ++a Data Structures Arrays Collections Type Definitions and Modifiers Class Reference type Supports inheritance Heap-allocated Struct Value type Cannot inherit Stack-allocated Copied by value Abstract class You can define common behavior in the abstract class (like Eat()), so derived classes don't have to re-implement it. Interface Defines method signatures without implementation Using: If classes are tightly coupled, changes in one class may affect others, leading to maintenance problems. C# does not support multiple inheritance of classes, which can limit the flexibility of your design. You want to handle objects of different classes in a uniform way, but the classes don't share a common base class. Sealed Cannot be overridden Operators and Methods == / Equals In C#, the == operator compares references for reference types (unless overloaded) and values for value types. The .Equals() method is used to compare values of objects and can be overridden in custom classes to provide a meaningful comparison based on the object's properties. Method overloading Method overloading in C# allows to define multiple methods with the same name but with different parameter Resource Management Using Statement When you use using, it makes sure that the object (like a file or database connection) is properly closed and cleaned up once you're done with it. You don't need to manually call .Dispose(). Database Concepts SQL